

CYBER ELIAS ACADEMY

COURSE SUMMARY

Cybersecurity

What You Will Learn in Every Session

9 SESSIONS

© 2026 Cyber Elias Academy · cea.ng

Cybersecurity — Course Summary

A quick reference of what is covered and what you will learn in each session.

QUICK NAVIGATION

Session 01 — Threat Landscape

Session 02 — Passwords MFA

Session 03 — Safe Browsing

Session 04 — Email Phishing

Session 05 — Device Hardening

Session 06 — WiFi Mobile Backups

Session 07 — Assessment Method

Session 08 — Improvement Plan Incident

Session 09 — Capstone Report

SESSION 01

Threat Landscape

Week 1 â€" Module 1: Hygiene (Stopâ€"Thinkâ€"Verify)

WHAT YOU WILL LEARN

- ✓ **Define** — cybersecurity in their own words and explain why it matters for ordinary people and small businesses.
- ✓ **Explain and apply** — the "Stopâ€"Thinkâ€"Verify" habit before clicking, paying, replying or sharing.
- ✓ **Name and describe** — the main malware families â€" virus, worm, trojan, spyware and ransomware â€" and give one plain example of each.
- ✓ **Define** — social engineering and phishing, and explain how they trick people (psychology, not just technology).
- ✓ **Describe** — common Nigerian scam and fraud patterns â€" fake bank SMS, WhatsApp impersonation, fake job offers, fake investment schemes â€" and list their usual red flags.
- ✓ **Distinguish correctly** — between the four core terms: **threat, vulnerability, exploit** and **risk**.
- ✓ **Explain** — why small businesses are attractive targets, using at least two reasons.
- ✓ **Identify red flags** — in five fictional scenarios and recommend the safe response for each, scoring at least 4 out of 5.

SESSION 02

Passwords MFA

Week 1 â€" Module 1: Hygiene (Stopâ€"Thinkâ€"Verify)

WHAT YOU WILL LEARN

- ✓ **Explain** — why passwords fail â€" guessing, reuse, breaches and weak patterns â€" in plain language.
- ✓ **Create** — a strong passphrase (length, uniqueness, no personal information) that is easy to remember and hard to guess.
- ✓ **Explain** — why reusing one password everywhere turns a single breach into many compromises.
- ✓ **Describe** — what a data breach is and **check** whether their **own** email appears in a known breach, using a public breach-checking service.
- ✓ **Set up** — multi-factor authentication (MFA) on their **own** account and explain the difference between authenticator-app MFA and SMS-code MFA.
- ✓ **Use** — a free password manager (Bitwarden) to generate and store a unique password, and explain how it protects them.
- ✓ **Describe** — account-recovery best practice (recovery email, recovery codes, honest security questions) and why it matters.
- ✓ **Produce** — a personal account-safety action list: one upgraded password, one MFA-enabled account, one password-manager entry.

SESSION 03

Safe Browsing

Week 1 â€” Module 1: Hygiene (Stopâ€“Thinkâ€“Verify)

WHAT YOU WILL LEARN

- ✓ **Explain** — what HTTPS/TLS does â€” it encrypts the connection between the browser and the website â€” and correctly state what the padlock does **and does not** mean.
- ✓ **Inspect a web address (URL)** — and identify the domain, the subdomain and the path, and spot suspicious elements (misspellings, extra words, unusual endings, numbers).
- ✓ **Recognise** — suspicious links, shortened URLs and urgency tricks, and decide whether a link is safe to click.
- ✓ **Choose safe download sources** — , avoid cracked/warez software, and check file types before opening.
- ✓ **Perform basic browser safety habits:** — keep the browser updated, manage extensions, and handle pop-up scareware without clicking.
- ✓ **Explain** — the risks of QR codes and public Wi-Fi, and outline safe behaviour for each.
- ✓ **Verify a website** — before entering personal or payment details.
- ✓ **Correctly classify** — at least 4 of 5 example web addresses as safe or unsafe, with a reason.

SESSION 04

Email Phishing

Week 2 â€” Module 2: Protecting People and Devices (theme: **Harden**)

WHAT YOU WILL LEARN

- ✓ **Describe in simple terms** — how an email travels from a sender to a recipient, naming the sender, the visible display name and the destination link as three things an attacker can fake.
- ✓ **Explain** — what phishing is and why it works, using the Stopâ€“Thinkâ€“Verify habit from Session 1.
- ✓ **Identify at least six red flags** — in a suspicious email, including look-alike domains, urgency, threats, unusual requests, generic greetings, spelling/grammar errors and mismatched link text versus destination.
- ✓ **Demonstrate "hover before you click"** — by moving the mouse over a link and reading the real destination without clicking it.
- ✓ **Check the true sender address** — rather than trusting the display name, and explain why the two can differ.
- ✓ **Distinguish phishing from spear-phishing and pretexting** — , and describe CEO fraud and fake-invoice scams in their own words.
- ✓ **Follow the correct response steps** — for a suspicious email: do not click, do not reply, report it, verify by a separate known channel, then delete/report as spam.
- ✓ **Connect email account protection** — to the work done in Session 2 â€” a unique strong password plus multi-factor authentication (MFA).
- ✓ **Assess five fictional phishing emails** — , correctly marking the red flags and the safe action for each.

SESSION 05

Device Hardening

Week 2 â€" Module 2: Protecting People and Devices (theme: **Harden**)

WHAT YOU WILL LEARN

- ✓ **Explain in simple terms** — what an operating-system update is and why leaving a device un-updated is dangerous (the "locked-door" idea).
- ✓ **Check Windows Update status** — on a Windows 10/11 device and confirm whether automatic updates are on.
- ✓ **Verify that Windows Defender real-time protection is on** — and **run a Quick scan**, then read the result correctly.
- ✓ **Confirm the Windows Firewall is on** — for the current network profile and explain what the firewall does in plain words.
- ✓ **Explain the difference between a standard user and an administrator account** — , and describe what a UAC prompt is asking.
- ✓ **Decide whether a piece of software comes from a safe source** — , and apply a rule for downloading and installing apps.
- ✓ **Manage startup programs** — so fewer unnecessary applications run automatically at boot.
- ✓ **Explain "defence in depth"** — â€" why antivirus alone is not enough and why several layers must work together.
- ✓ **Produce a short "device hardening checklist"** — about a device they are authorised to use.

SESSION 06

WiFi Mobile Backups

Week 2 â€" Module 2: Protecting People and Devices (theme: **Harden**) â€" final session of Week 2

WHAT YOU WILL LEARN

- ✓ **Explain** — why Wi-Fi encryption (WPA2/WPA3) matters and describe what happens to data on an open network.
- ✓ **List and explain** — the key router hardening steps: change the default admin password, set a strong Wi-Fi password, turn off WPS, and enable a guest network.
- ✓ **Describe the risks of public Wi-Fi** — and state what they should avoid doing on it.
- ✓ **Apply mobile safety basics** — : screen lock, reviewing app permissions, updating apps, using only official app stores, enabling find-my-device, and avoiding sideloaded/cracked apps.
- ✓ **Explain the 3-2-1 backup rule** — in simple words: 3 copies, 2 different media, 1 offsite.
- ✓ **Decide what is worth backing up** — and compare cloud, external drive and USB options with their trade-offs.
- ✓ **Explain why a backup must be tested** — ("a backup you have never restored is not a backup") and describe how to test a restore.
- ✓ **Build a backup plan** — for a small business and **perform a folder backup to an external drive/USB and restore one file**.

SESSION 07

Assessment Method

Week 3 â€" Module 3: Assess and Improve

WHAT YOU WILL LEARN

- ✓ **Define** — a defensive security assessment in their own words and explain how it differs from hacking or penetration testing.
- ✓ **State the rule of authorisation** — â€" that a system may only be assessed with the owner's written permission â€" and give one example of an authorised target and one of an unauthorised target.
- ✓ **Build a written checklist** — covering at least seven areas of a small business: accounts and passwords, MFA, device updates, antivirus and firewall, Wi-Fi, backups, staff awareness and phishing risk, and data handling.
- ✓ **Gather evidence safely** — by describing and recording what is seen, without changing anything and without testing other people's systems.
- ✓ **Rate a finding's risk** — using likelihood Ã— impact and assign Low, Medium or High.
- ✓ **Prioritise findings** — and justify which problem should be fixed first.
- ✓ **Produce a findings table** — with columns for the issue, the risk level, and the reason, for a fictional business scenario.

SESSION 08

Improvement Plan Incident

Week 3 â€" Module 3: Assess and Improve

WHAT YOU WILL LEARN

- ✓ **Convert** — a findings table into a prioritised improvement plan that separates **quick wins** from **longer projects**.
- ✓ **Write a clear recommendation** — using the five questions: what, why, how, who, when.
- ✓ **Give cost-aware advice** — to a small Nigerian business, choosing free or cheap fixes before expensive ones.
- ✓ **Describe the six basic incident-response steps** — â€" Prepare, Identify, Contain, Eradicate, Recover, Learn â€" in simple words.
- ✓ **Explain the first-hour actions** — for a suspected breach or ransomware: disconnect from the network, do not pay blindly, preserve evidence, change passwords from a clean device, inform the right people, and restore from a clean backup.
- ✓ **Produce a one-page incident response card** — that a non-technical staff member could follow.
- ✓ **Take part in a tabletop incident drill** — , making sensible decisions in the correct order.

SESSION 09

Capstone Report

Week 3 â€” Module 3: Assess and Improve (capstone build and present)

WHAT YOU WILL LEARN

- ✓ **Assemble** — a complete defensive security assessment report for a fictional, authorised business, containing all required sections.
- ✓ **Justify** — the scope and authorisation statement, and demonstrate the ethics of assessing only permitted systems.
- ✓ **Present** — their findings table, risk ratings, and prioritised improvement plan clearly to a non-technical audience.
- ✓ **Defend their decisions** — in a short viva, answering questions about risk ratings and recommendations.
- ✓ **Demonstrate** — one hardened, defensive action live (e.g., showing MFA switched on, or a backup completed) on their own academy account/device.
- ✓ **Explain honestly** — what this qualification does and does not make them: a safer user and a junior IT-support beginner with defensive awareness â€” **not** a hacker or penetration tester.